Computer Science & IT

C Programming



Functions & storage classes

Lecture No. 02



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Recap of Previous Lecture



Topic — Function , prototype , parameter & arguments

Topic — Function Declaration

Topic — Function definition

Topic — Function call

Topic

Slide

Topics to be Covered



- Topic — Activation tree
- Topic — Problem practice
- Topic — Recursion
- Topic
- Topic

function / Activation Record / : The most important topic.

Slide

#Q Consider the following program

```
#include <stdio.h>
void a(){printf("1");}
void b(){a();printf("2");}
void c(){a();b();printf("3");}
void d(){a();b();c();printf("4");}

int main(){
   d();
}
```

program?

(A) 11211234

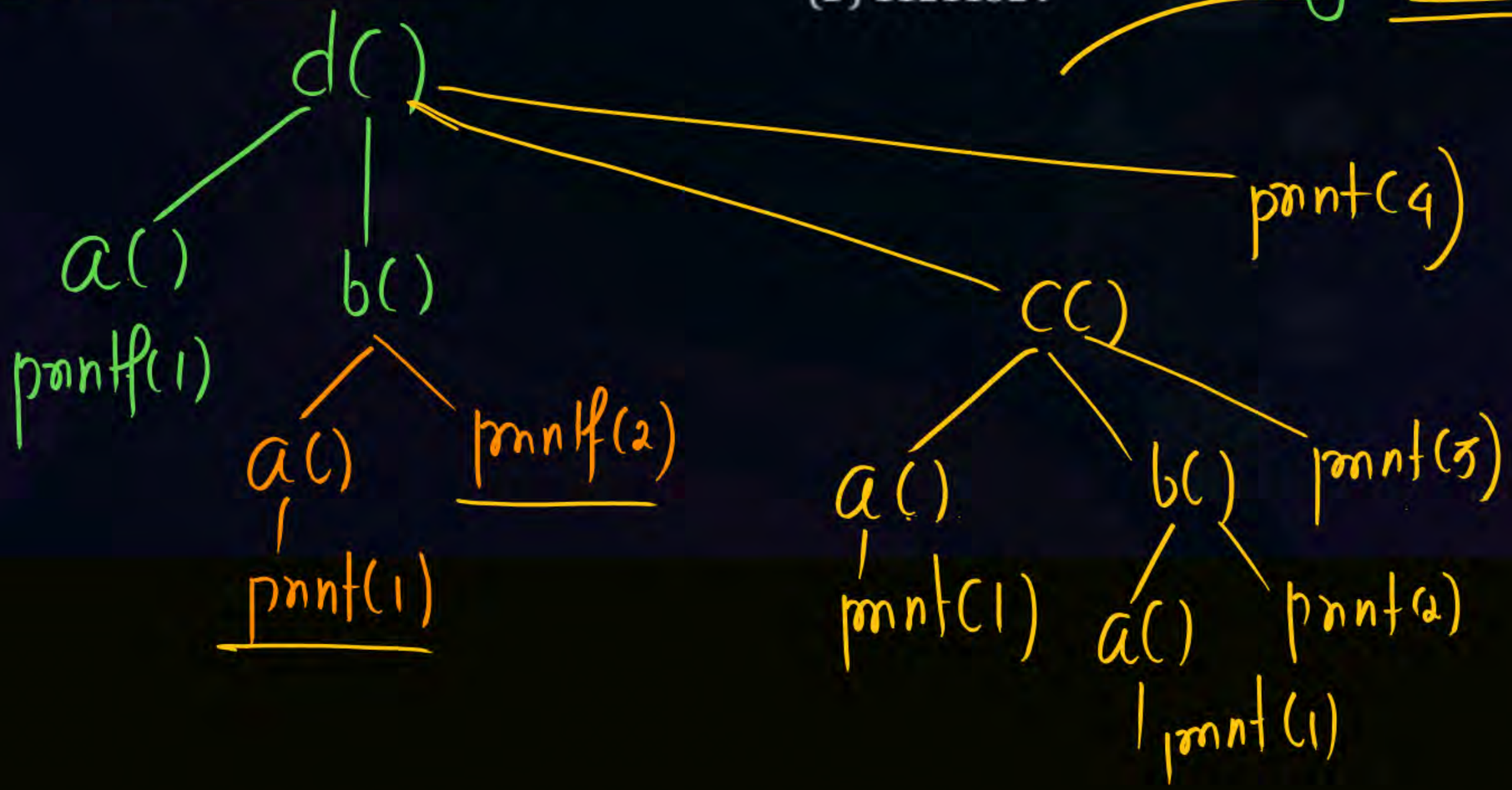
(B) 11211243

(C) 11212134

(D) 11211324

1 1 2 1 1 2 3 4

manage tree



Slide



GATE 2017

r = x

#Q

Consider the C program fragment below which is meant to divide x by y using repeated subtractions. The variables x, y, q and r are all unsigned int.

```
while (r >= y)    {
r = r - y;
q = q +1;
}
```

repeated

Which of the following conditions on the variables x, y, q and r before the execution of the fragment will ensure that the loop terminates in a state satisfying the condition x== (y*q + r)?

(A) (q==r) && (r==0)

(B) (x>0) && (r==x) && (y>0)

(C) (q==0) && (r==x) && (y>0)

(D) (q==0) && (y>0)

$\frac{X}{Y}$

divisor cant

repeated Substraction

q → quotient

r - remainder

X == (y*q+r)

Slide

7/2 dividend = 7 divisor = 2 q = 0 , r =

~~int~~ cherahan 1 7 - 2 = 5 q = 1 q = 3

5 - 2 = 3 q = 2 7/2 = 3

3 - 2 = 1 q = 3

r = X

Y

q = 0

r = 1

$10/4$ $q=2, r=2$

$r=10$, $q=0$ y

$r=2$

$q=2$

$r-y$

$10-4=6$ $\quad q=q+1=1$

$r=6-4=2$ $\quad q=q+1=2$

$r=2$

#Q

Consider the following program

```
#include<stdio.h>
void a();
void b();
void c();

int main()
{
    a();
    b();
}

void a(){printf("a"); b();}
void b(){printf("b"); c();}
void c(){printf("c");}
```

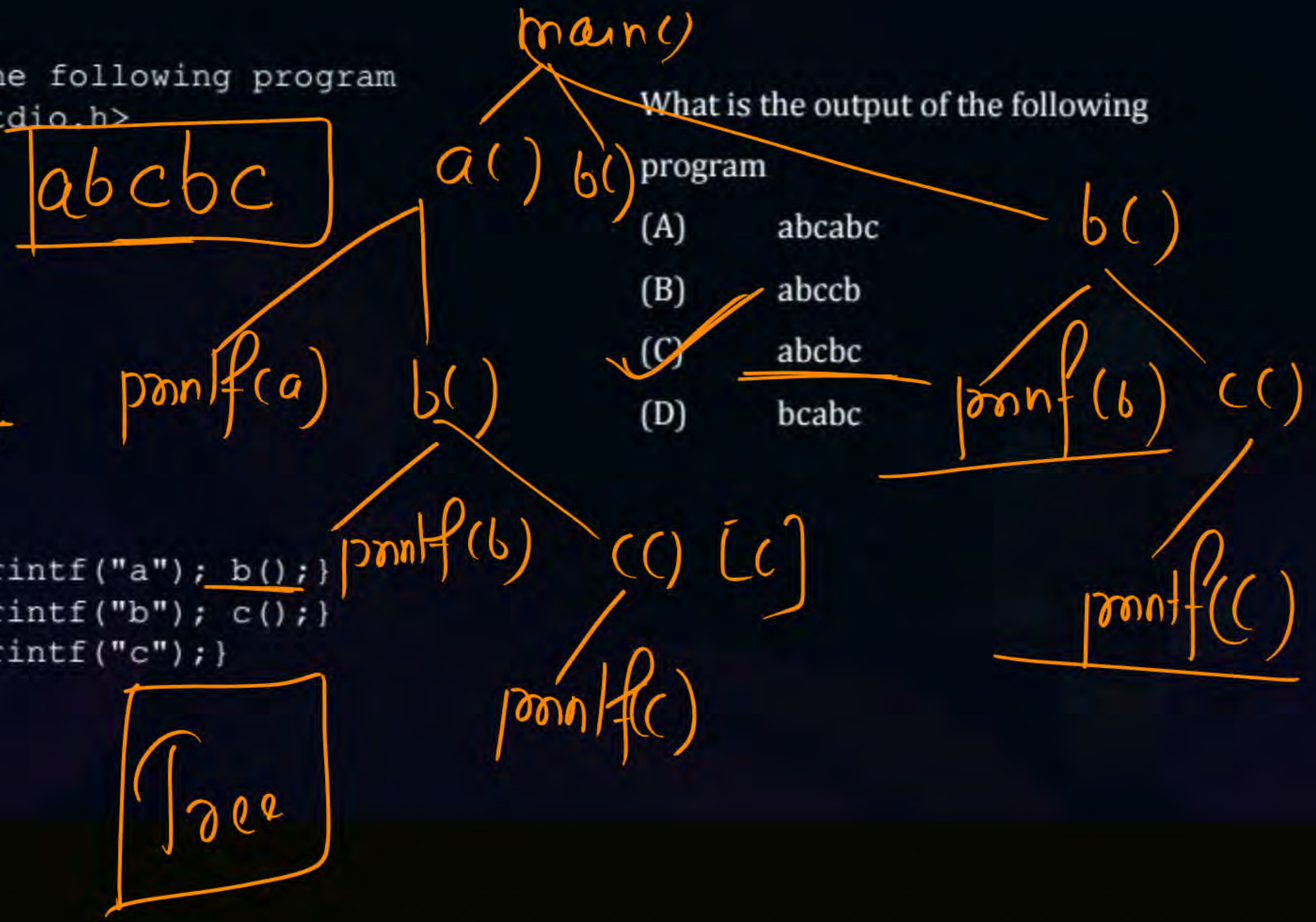
What is the output of the following program

(A) abcabc

(B) abccb

(C) abcbc

(D) bcabc



Slide

Consider the following C program. Assume parameters to a function are evaluated from right to left.

```
#include <stdio.h>
int g(int p) { printf("%d", p); return p; }
int h(int q) { printf("%d", q); return q; }
void f(int x, int y) {
    g(x);
    h(y);
}
int main() {
    f(g(10), h(20));
}
```

Which one of the following options is the CORRECT output of the above C program?

(A) 20101020

(B) 10202010

(C) 20102010

(D) 10201020

Imp

10201020

20101020

Clear Hua

20 10 10

Slide

f(g(10), f(20)) what is the order if right to left

Not given

order in C Language Not defined



GATE 2005

#Q

```
double foo (double);                /* Line 1 */
int main() {
    double da, db;
    //input da
    db = foo(da);
}
double foo (double a) {
    return a;
}
```

int

if Line 1 declaration

Error/warning

foo return type int

f(x+y, z++ + ++z)

f(int, int)

The above code compiled without any error or warning. If Line 1 is deleted, the above code will show:

mismatch

A. no compile warning or error

B. some compiler-warnings not leading to unintended results

C. some compiler-warnings due to type-mismatch eventually leading to unintended results

D. compiler errors

Slide



GATE 2015 Set-3

#Q

Consider the following C program:

```
#include<stdio.h>
int f1(void);
int f2(void);
int f3(void);
int x=10;
int main()
{
    int x=1;
    x += f1() + f2() + f3() + f2();
    printf("%d", x);
    return 0;
}
```

```
int f1() { int x = 25; x++; return x;}
int f2() {int x = 50; x++; return x;}
int f3() { x *= 10; return x;}
```

The output of the program is _____.

global

$X = x + f_1() + f_2() + f_3() + f_2()$

$1 + 26 + 51 + 100 + 51$

229

x += y

x = x + y

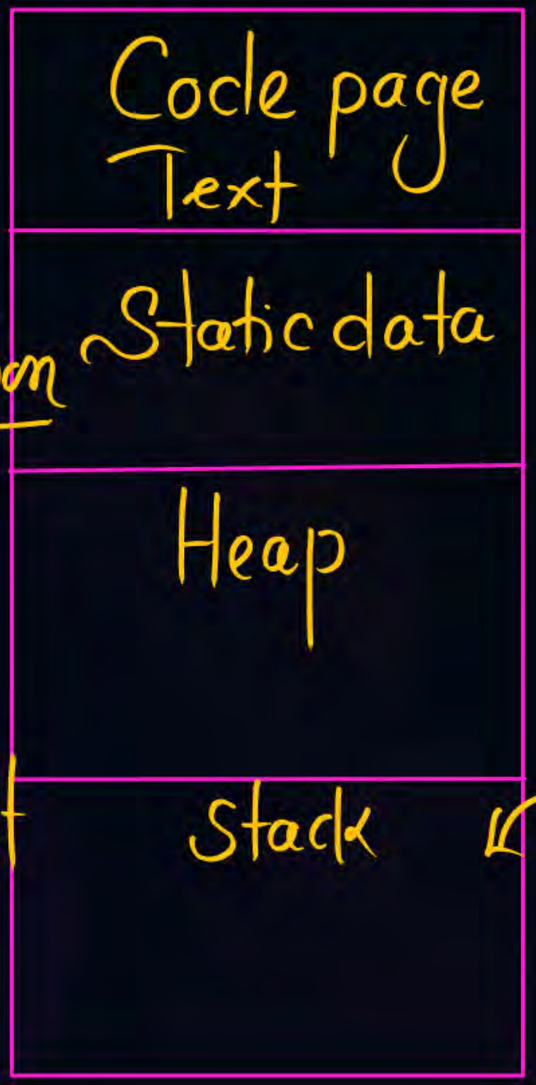
Slide

Activation Tree and Records



when a program Run

Memory



Stack data structure

LIFO

Last in first out

Run time Stack

Activation Record

Slide



Recursion

```
#inalcude<stdio.h>
void print(int n){
     if (n <= 0)  return;
     printf("%d", n);
     print(n-1);
}
int main(){
     print(5);
     return 0;
}
```



Recursion

```
#include<stdio.h>
void print(int n)    {
     if (n <= 0)  return;
     print(n-1);
     printf("%d", n);
     }

int main(){
     print(5);
     return 0;
}
```

Slide



Recursion

```
#include<stdio.h>
void print(int n)    {
     if (n <= 0)  return;
     print(n-1);
          printf("%d", n);
          print(n-1);
     }
int main(){
          print(3);
          return 0;
}
```



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Slide

THANK - YOU

